

NEURAL BELLMAN OPERATORS: RESPONSE TO THE R15 REFEREE REPORT

QIAN QI

Peking University

1. OVERVIEW AND THE REVISED CONTRIBUTION

We thank the referee for separating the progress in economic verification from the still-demanding comparison of continuation learning with credible alternatives. We have revised the original NBO article rather than substituting a different research topic. The new main text centers on the learned continuation object, feasible improvement, a signed Bellman mechanism, and the economic value of reuse. The original mathematical results and economic applications remain in the current supplement, with exact preceding sources preserved separately.

Three completed additions materially change the evidentiary position. First, a prospective signed occupation assessment supplies positive continuous-economy gain lower endpoints in both dimensions where the original norm-based sufficient account was negative. Second, an independent confirmation bank, evaluated only after the paired theorem and its constants were fixed, reproduces the specified NBO–HJB comparison. Third, an exact decision-value proposition translates fixed-dictionary gradient compression into expected economic payoff in a precisely stated quadratic specialization. Its proof identifies removed economically weighted variance and approximation cost, and records the feasibility condition under which the equality holds.

The two broader registered empirical families are also complete. The five-method continuation menu retains all 128 trials, 640 method fits, 216 primary events, and 128 scalar statements. The strengthened-HJB study retains four economic designs, 64 trials, 320 policies, and 648 events. Their inclusion changes this from a design-only response into an outcome-level comparison. NBO has material advantages over fitted alternatives in difficult reusable-continuation cases and practical equivalence with cached sample-average optimization at lower construction and query work in six of eight final-stage cases. The sample-average method wins materially in

both long-continuation cases, and strengthening HJB leaves most direct contrasts unresolved. All outcomes, not only favorable signs, enter the revised article and complete supplement.

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2. M1 AND B1: THE LEARNED CRITIC’S INCREMENTAL ECONOMIC VALUE

Referee concern. Positive verified payoff for the NBO package does not establish that fitting the critic improves on cached Raw continuation targets. The reported Raw procedure has similar payoff, lower work, and unresolved direct intervals.

Revision. Section 6 states the precise common-continuation object and the valid forms of reuse. A payoff or derivative observation is reusable at its argument when future primitives and the reference policy remain fixed. Innovations can also be reused at another argument, but the state recursion and payoff then have to be reevaluated. This prevents a comparison that artificially charges Raw for unnecessary resimulation while treating NBO reuse as free.

Proposition 6 gives an exact bias–variance identity for a scalar-gradient dictionary fixed independently of the fitting labels. Proposition 7 then gives its economic counterpart. In the stated quadratic Bellman tasks, the expected weighted payoff advantage equals one half of removed economically weighted noise variance minus approximation cost. Correct representation and nonzero removable economic noise give strict improvement. The statement also requires that the optimizers be feasible interior solutions; it does not replace a constrained nonlinear actor by an unconstrained first-order formula. The proof is complete, and independent finite-support manufactured tests check both favorable and adverse approximation regimes.

The completed nonlinear comparison is presented in Section 7.13, with its design and every registered outcome in Appendix R. A vector costate fit, cached Raw actor, task-conditioned direct actor, and Raw sample-average optimizer receive

legitimate reuse as well. All stages and all declared economic designs are retained. The fixed-dictionary proposition gives a proved economic mechanism; the nonlinear outcomes are independently evaluated rather than identified with that proposition.

Completed evidence and scope. The original and fresh-bank NBO–Raw intervals remain visible and unresolved. A Raw method applying the same projection has the same estimator and does not lose merely because its name is Raw. The complete menu now supplies a direct incremental comparison. In the long fifty-dimensional case, the NBO advantage over the vector costate, Raw actor, and direct actor has lower endpoints 16.38, 18.02, and 3.27 in units of 10^{-4} . The untouched intermediate fifty-dimensional case also yields material Raw-actor and direct-actor advantages. Across all 24 registered stage–economy cases, the numbers of positive lower endpoints against vector, Raw actor, direct actor, and cached sample-average optimization are 22, 19, 16, and zero. Practical equivalence with the sample-average method holds in six of eight final-stage cases at the fixed margin, while the two long cases show material inferiority. These results identify a scoped economic tradeoff rather than necessity of a critic in every implementation.

Binding-constraint extension. Proposition [S.16](#) supplies a complete proof of the decision-value identity on stable active faces. It allows each task to have binding constraints when their tangent spaces jointly span the continuation coordinates. The active faces and scalar-gradient dictionary are fixed for the original centered observation law; no post-selection recentering is allowed. Ten additional finite-law tests compare the identity with exact constrained KKT solutions, including off-diagonal curvature, unequal economic weights, misspecification, and explicit rejection of face switching and singular metrics. These manufactured checks are distinct from the registered economic outcomes.

3. M2 AND B2: A STRONGER NEURAL HJB COMPARATOR

Referee concern. The original noisy residual product, restricted fitting envelope, and costly bisection deployment make its HJB implementation too narrow a basis for a broad method ranking.

Revision. The new baseline specification uses exact architecture-specific covariance Hessian contraction in both dimensions. Independent automatic differentiation checks the contraction and its weight gradients. The prospective fitting family crosses two collocation laws and two learning rates, allows up to 1,200 Adam updates and a subsequent L-BFGS stage, and selects only by exact residual RMS on independent selection banks. Separate untouched residual and probe-variance audits do not select a model. The code preserves failures, deterministic tie rules, and every retained candidate.

Two deployments are prespecified: a safeguarded Newton greedy solve, with charged fallback, and a distilled width-32 actor trained on cached greedy actions. Both receive final common-path evaluation; neither is selected after inspecting its economic payoff. The full shared HJB construction cost is charged to each reported deployment and disclosed as shared work. Appendix R gives the full economic and computational specification.

Completed evidence and scope. The current original and replicated superiority statements are expressly limited to the original specified HJB implementation. They are not relabeled as evidence against the strengthened family. The strengthened family is now evaluated completely in Section 7.14. The low-volatility fifty-dimensional NBO-minus-greedy and NBO-minus-distilled intervals are contained in $[0.000334, 0.000697]$ and $[0.000340, 0.000704]$. The other fourteen HJB contrasts include zero. In particular, the enlarged family substantially narrows the original comparator gap; its point-estimate reversals are retained, not hidden. All four candidate fits, independent residual diagnostics, both deployments, and complete work remain in the evidence tree. Sampled residuals and successive-iterate differences are not called uniform HJB error certificates.

4. M3 AND B3: PROSPECTIVE CONFIRMATION OF PAIRED TRANSFER

Referee concern. A valid post-freeze deterministic refinement on the old bank is not the same as a prospectively specified evaluation.

Revision and completed evidence. Section 7.7 now presents a separate new-bank confirmation. All 128 original selected policies are retained without retraining. Each

of sixteen original streams has 8,192 independent paths on 2,048 cells. The paired theorem, constants, code, bank identity, confidence family, and economic margin precede those observations. The new family has its own 0.01 budget and 238 events.

The fresh paired NBO-minus-HJB intervals are contained in $[0.000176, 0.000415]$ and $[0.001342, 0.001653]$. Both lower endpoints exceed 10^{-4} . The original ten-dimensional interval $[-0.000073, 0.000663]$, the later old-bank refined interval $[0.000179, 0.000411]$, and the fresh interval remain different entries with their own chronology. No policy, old sample, historical stopping decision, or old confidence allocation has been changed.

This closes the request for independent prospective confirmation of the fixed-policy comparison. It does not claim new training streams or a second economic calibration; those are separate parts of the registered robustness design.

5. M4, B1 AND B6: RAW, DIRECT POLICY, AND COMPARATIVE WORK

Referee concern. Intervals containing zero do not establish equivalence, and the original work data do not identify an NBO advantage over Raw.

Revision. The introduction, economic interpretation, fresh-bank section, and conclusion now state the direct-comparison object explicitly. Every original Raw and direct-policy result is retained. The fresh comparisons with both methods in both dimensions include zero and extend outside the $\pm 10^{-4}$ band. They therefore establish none of superiority, equivalence, or noninferiority.

The work account preserves the original fifty-dimensional means: approximately 357.80 seconds for NBO, 346.22 for Raw, 596.94 for direct policy, and 1690.10 for the original HJB procedure. Raw remains cheaper than NBO in that design. We do not convert NBO's advantage over the specified direct-policy work account into proof that its critic is necessary. Unattained targets, failed checks, shared work, and final confirmation remain charged.

The common-continuation section adds a transparent amortization calculation. A lower per-task cost repays additional shared fitting work only beyond its explicit break-even task count. This is an accounting implication, not a proof of economic noninferiority. The completed menu measures construction, deployment, optimiza-

tion, and the full shared confirmation bundle for all five methods and every stage. NBO has lower observed complete third-stage construction and query work than cached sample-average optimization in all eight cases and satisfies the prespecified practical-equivalence criterion in six. For the quarterly and untouched intermediate fifty-dimensional cases, the construction/query clocks are 14.34 versus 62.90 seconds and 12.70 versus 79.46 seconds. Both long cases instead have material sample-average payoff advantages. Raw actors remain cheaper than NBO. The shared confirmation bundle is retained separately and can be charged in full to either method; it is not divided into fictional standalone costs. Consequently the claim is a measured work–accuracy tradeoff, not universal dominance or an invented work-to-certified-gain stopping time.

6. M5 AND B4: ECONOMIC ROBUSTNESS AND TRAINING RANDOMNESS

Referee concern. Sixteen declared streams at a pilot-selected calibration are not evidence of general economic or optimizer-law robustness.

Revision. All method-level statements retain the finite declared stream estimand. Their empirical distribution is fully enumerated, not identified with a universal population of optimizers or random seeds. Independent confirmation of old policies is separately named as such.

The strengthened-HJB robustness design fixes four economic cases before new fitting and confirmation: historical high volatility, historical low volatility, a longer horizon with intermediate volatility, and a joint stress of coupling, terminal dispersion preference, feasible action radius, and initial heterogeneity. It uses eight new complete training streams in each dimension. The joint stress is not assigned a single-primitive causal interpretation.

The continuation-menu design independently includes the original low-volatility anchor, the disclosed reuse pilot, a longer unpiloted continuation, and untouched intermediate primitives. The protocol retains all four designs and all work stages, rather than selecting the design with the most favorable final comparison. The completed results in this edition do not silently substitute seed variation for economic variation. The complete outcome-level robustness family is now included: all eight

NBO gain intervals have positive lower endpoints, while the Raw contrasts and most strengthened-HJB contrasts remain unresolved. The menu likewise retains all four designs, including the untouched intermediate case, all 1,920 stage candidates, and every unfavorable comparison. These are genuine variations in economic primitives, reported separately; the estimands still do not extrapolate to an unobserved universal optimizer distribution.

7. M6 AND B5: ABSOLUTE ECONOMIC ACCURACY

Referee concern. The broad adapted-control regret bounds are much larger than the economic decision margin, and the scalar classical comparison does not support universal neural superiority.

Revision. We retain the broad adapted-control bounds and consistently distinguish them from a 10^{-4} near-optimality certificate. The original NBO bounds remain approximately 0.025970104 and 0.036842066 before outward rounding. The new bank yields upward-rounded bounds 0.025971 and 0.036845. Neither calculation is represented as a tight high-dimensional HJB solution.

The original scalar Howard results, the classical policy's higher population payoff, the state-specific NBO/direct-policy reversal, domain and grid changes, and the full-box/tube comparison remain in the current manuscript and supplement. Their finite-grid or scalar-model scope is unchanged.

Proposition S.15 adds a different absolute accuracy object. Its variation bounds give global curvature on an entire prespecified scalar intervention segment. An independent derivative interval then bounds the best action over every point of that segment, including unevaluated points. The support formula remains valid when the strong-concavity coefficient is nonpositive. The derivative interval is derived from a protected paired secant and its deterministic curvature allowance, not from treating an automatic derivative as an exact expectation. The completed scalar assessment now supplies numerical whole-segment accuracy, not only a route to it. Eighty of the 128 prespecified candidates meet the 10^{-4} margin, including every quarterly fifty-dimensional stream with upward-rounded gap bounds no larger than 0.0000141. All forty-eight nonattainments remain displayed; the largest

long fifty-dimensional bound is 0.1963168. The protected secant and curvature are unchanged from their pre-confirmation specification. This is not mislabeled as a full adapted-control or continuous-time guarantee.

8. M7 AND B7: A POSITIVE AND PRECISE MECHANISM ACCOUNT

Referee concern. The old occupation-risk and maximization-gap endpoints are too conservative to establish positive mechanism welfare or a critic advantage over Raw.

Revision and completed evidence. Proposition 4 and the complete construction in Proposition S.12 replace subtraction of separate marginal uncertainty bounds by a shared signed observation. For every draw, $G = M - C$ before clipping and statistical inference. Its expectation is the finite occupation Bellman gain. Antithetic endpoint differences remove a known terminal innovation term; the remaining range uses economic coefficients and the action tube rather than a fitted network norm. Holding and continuous-payoff transfer are charged after the finite lower endpoint.

The new assessment retains every one of the 32 original NBO policies and uses 1,024 fresh occupation bridges per stream, for 32,768 bridges in total. The six simultaneous two-sided statements receive a new 0.01 family budget. The signed continuous-economy lower endpoints are at least 0.000767 in dimension ten and 0.000776 in dimension fifty. The full stream table, range, tail, arithmetic, prefix, holding, transfer, and work accounts remain available.

The original negative norm-based sufficient bounds are not removed. The new result explains why their negativity need not imply a negative realized Bellman gain. At the same time, cancellation of the learned value in the final verification statistic does not identify its incremental value in candidate construction. Critic-versus-Raw risk intervals remain separate; no risk superiority is deduced from a positive signed policy gain.

9. M8: CHRONOLOGY, SELECTION, AND INDEPENDENT REPLICATION

The article distinguishes original observations, same-event deterministic refinement, and fresh prospective confirmation. The generating source and candidate

identities are immutable. New confirmation does not repeat fitting or retrospectively change an online stop. The strengthened HJB family uses residual-only selection before final payoff. The menu fixes the economic designs, stages, query law, task law, methods, sample sizes, and 10^{-4} margin before its confirmation. Disclosed development pilots and failed source-closure rehearsals are preserved rather than rewritten as prospective observations.

The new confidence allocation has four disjoint named budgets totaling 0.05 by a union bound. No unused allocation is recycled, and historical families are not silently included in this new union. The reproduction record provides each family's distinct source, outcome status, and estimand. All four named registered families are now complete. The publication record binds each to its distinct pre-confirmation source and immutable result tree. This establishes the chronology of the executed studies without making any claim about every possible unregistered economic or optimizer population.

10. M9 AND B8: A COHERENT NBO ARTICLE WITHOUT LOSS OF CONTENT

The main article now follows the operator, capital economy, signed mechanism, common continuation, and economic evidence. General approximation and selection theorems, detailed inherited theory, fee accounting, and the broader application discussion are placed in subject-based sections of the current supplement. Their original labelled statements and complete proofs are retained. Recursive utility, endogenous preference formation, temporal selves, and games are not deleted, replaced by another topic, or reduced to inaccessible historical references.

The revised abstract and introduction distinguish theorem scope, policy-specific welfare, comparative method evidence, and implementation guarantees. The source branch and exact old roots remain available for a textual comparison. The response and reproduction files, rather than the body of the article, carry commit identities and execution chronology. Existing adverse findings are retained in the economic interpretation and full tables, not hidden solely in an archive.

11. M10: DECISION AND REPORTING SEMANTICS

Every displayed inference is identified by its economic model, information set, feasible controller, comparison object, and confidence family. Finite stream averages, conditional fixed-policy statements, and broad adapted-control bounds are not conflated. Stored finite-period menu coefficients are exact inputs to that finite economy; they are not automatically exact continuous-time coefficients. One antithetic pair is one independent observation, not two independent paths.

Historical online attainment remains different from final confirmation. A restricted-mean censoring convention is not actual elapsed work. Residual stopping in the strengthened HJB design is not an economic work-to-certified-gain stop. A scalar whole-segment optimality bound is not a full high-dimensional certificate. A confidence interval containing zero is not an equivalence decision. These distinctions are reflected in the abstract, introduction, theory statements, table notes, and conclusion.

12. PRESERVATION, VERIFICATION, AND REMAINING COMPARATIVE SCOPE

The reading edition preserves the reviewed source tree apart from the three new reading roots, whose exact preceding bytes have archive destinations. Historical numerical files and scientific sources are not edited. The published preservation and compilation checks bind the reading edition to its parent commits and require resolvable references. Local manufactured checks separately cover the signed construction, menu statistics and transport, exact HJB derivatives, paired capital transfer, and the new decision-value identity; such checks are not presented as economic outcome evidence.

The substantive completed changes are the signed positive occupation mechanism, prospective paired replication, decision-value theorem, protected conditional comparison and scalar-accuracy theory, full five-method continuation evidence, full strengthened-HJB cross-calibration evidence, and the reorganized article and supplement. Every one of the four registered evidence families is complete. The conditional Raw, work-accuracy, scalar, and HJB conclusions now rest on the complete outcomes rather than plans. Unresolved comparisons and broad full-adapted-control gap bounds remain honestly stated; they do not erase the positive economic

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